

HOSPITAL MANAGEMENT SYSTEM

Project Plan

Timeline, Resource Allocation, and Task Breakdown

Project Title	Hospital Management System
Project Duration	2 Months (8 Weeks)
Start Date	Week 1, Month 1 (March 2026)
End Date	Week 8, Month 2 (April 2026)
Team Size	2 Members
Member 1	Syed Ukkasha Ahmed
Member 2	Yesaullah Sheikh
Version	1.0

1. Project Overview

1.1 Project Description

The Hospital Management System (HMS) is a Flask-based web application that digitises and integrates the core operational workflows of a hospital. The system serves five user roles: Doctor, Nurse, Receptionist, Store Manager, and Patient. Key functional areas include patient registration, appointment scheduling, inpatient admission and discharge, nurse-managed vitals tracking, prescription and medication management, inventory and supply chain management, and billing.

1.2 Project Objectives

- Deliver a fully functional, role-based hospital management web application within 8 weeks
- Complete all core modules: Authentication, Patient Management, Appointment Management, Inpatient Workflow, Prescriptions, Billing, and Inventory
- Produce all required project documents: SRS, Design Document, and this Project Plan
- Conduct unit and integration testing across all modules before final submission
- Deploy the system in a production-ready configuration using Gunicorn

1.3 Team and Roles

Team Member	ID	Primary Responsibilities	Secondary Responsibilities
Member 1	M1	Backend: Models, Authentication, Doctor & Patient blueprints, SRS documentation	Database migrations, integration testing, deployment
Member 2	M2	Backend: Receptionist, Nurse, Store blueprints, Design documentation	Frontend templates, billing logic, unit testing

2. Work Breakdown Structure (WBS)

The project is decomposed into 7 phases. Each phase contains specific tasks with assigned owners, estimated effort, and dependencies.

Task ID	Task Name	Owner	Duration	Effort (hrs)	Dependencies	Deliverable
1.0	PHASE 1: Project Initiation & Planning	M1 + M2	Week 1	-	-	-
1.1	Review project requirements and existing codebase	M1 + M2	Days 1-2	8	None	Codebase understanding notes
1.2	Draft SRS document structure and assign sections	M1	Days 2-3	6	1.1	SRS Draft v0.1
1.3	Set up development environment and Git repository	M2	Day 1	4	None	Dev environment ready
1.4	Define project plan and create Gantt chart	M1 + M2	Days 3-5	6	1.1	Project Plan v1.0
1.5	Finalise SRS document with all sections	M1 + M2	Days 4-7	12	1.2	SRS v1.0
2.0	PHASE 2: Database & Core Architecture	M1 + M2	Week 1-2	-	-	-
2.1	Finalize all SQLAlchemy models (models.py)	M1	Days 3-5	8	1.1	models.py final
2.2	Create and run initial Alembic migration	M1	Day 5	3	2.1	DB schema v1
2.3	Implement Application Factory (app/__init__.py)	M2	Days 3-4	4	2.1	create_app() tested
2.4	Seed room data (seed_rooms.py)	M2	Day 6	2	2.2	Rooms seeded in DB
2.5	Implement authentication blueprint (auth/routes.py)	M1	Days 6-7	8	2.3	Login/Register working

Task ID	Task Name	Owner	Duration	Effort (hrs)	Dependencies	Deliverable
3.0	PHASE 3: Core Module Development	M1 + M2	Weeks 2-4	-	-	-
3.1	Implement Patient blueprint (CRUD + medical records)	M1	Week 2 D1-4	16	2.5	Patients module complete
3.2	Implement Doctor blueprint (appointments, prescriptions)	M1	Week 2 D4 - Week 3 D2	20	3.1	Doctors module complete
3.3	Implement Nurse blueprint (assignments, vitals, rooms)	M2	Week 2 D1-5	18	2.5	Nurses module complete
3.4	Implement Receptionist blueprint Part 1: patients, appointments	M2	Week 3 D1-3	16	3.1	Receptionist Part 1 done
3.5	Implement Receptionist blueprint Part 2: admission, discharge, billing	M2	Week 3 D3-5	20	3.4, 3.3	Receptionist full module done
3.6	Implement Store blueprint (inventory, suppliers, orders)	M1	Week 3 D3 - Week 4 D2	18	2.5	Store module complete
4.0	PHASE 4: Frontend Templates	M1 + M2	Weeks 3-5	-	-	-
4.1	Create base.html with navigation and flash messages	M2	Week 3 D1	4	2.5	base.html complete
4.2	Auth templates (login.html, register.html)	M2	Week 3 D1-2	4	4.1	Auth UI complete
4.3	Doctor dashboard and patient detail templates	M1	Week 3 D3-5	8	3.2, 4.1	Doctor UI complete
4.4	Nurse dashboard, patient status, rooms templates	M2	Week 4 D1-3	8	3.3, 4.1	Nurse UI complete

Task ID	Task Name	Owner	Duration	Effort (hrs)	Dependencies	Deliverable
4.5	Receptionist templates (admission, billing, rooms)	M2	Week 4 D3-5	12	3.5, 4.1	Receptionist UI complete
4.6	Store templates (inventory, orders, suppliers)	M1	Week 4 D4 - Week 5 D1	8	3.6, 4.1	Store UI complete
4.7	Patient-facing templates	M1	Week 5 D1-2	6	3.1, 4.1	Patient UI complete
5.0	PHASE 5: Integration & Testing	M1 + M2	Weeks 5-6	-	-	-
5.1	Integration testing: authentication and RBAC guards	M1	Week 5 D2-3	8	Phase 3 done	RBAC test report
5.2	Integration testing: patient admission to discharge flow	M2	Week 5 D3-4	8	3.5	Admission flow test report
5.3	Integration testing: prescription and dispensing	M1	Week 5 D4-5	6	3.2, 3.6	Prescription test report
5.4	Integration testing: billing (inpatient + outpatient)	M2	Week 6 D1-2	6	3.5	Billing test report
5.5	Integration testing: store inventory and order workflow	M1	Week 6 D2-3	6	3.6	Store test report
5.6	Bug fixing and regression testing	M1 + M2	Week 6 D3-5	12	5.1-5.5	Bug fix log; clean test run
6.0	PHASE 6: Documentation	M1 + M2	Weeks 1-7	-	-	-
6.1	SRS Document (delivered in Phase 1)	M1	Week 1	12	-	SRS v1.0
6.2	System Architecture & Design Document	M2	Weeks 2-3	12	2.1, 3.0	Design Document v1.0

Task ID	Task Name	Owner	Duration	Effort (hrs)	Dependencies	Deliverable
6.3	Project Plan (this document)	M1 + M2	Week 1	6	-	Project Plan v1.0
6.4	User Manual (role-by-role)	M2	Weeks 6-7	10	Phase 4 done	User Manual v1.0
6.5	README and code inline documentation	M1	Week 7	6	Phase 5 done	README + docstrings
7.0	PHASE 7: Deployment & Final Review	M1 + M2	Week 7-8	-	-	-
7.1	Configure Gunicorn and production environment variables	M1	Week 7 D1-2	4	Phase 5 done	Production config ready
7.2	Final system review and demonstration preparation	M1 + M2	Week 7 D3-5	8	7.1	Demo-ready system
7.3	Final documentation review and packaging	M2	Week 8 D1-3	8	6.1-6.5	Final document package
7.4	Project submission	M1 + M2	Week 8 D5	2	7.1-7.3	Final submission

3. Gantt Chart — 8-Week Timeline

The chart below shows task durations across 8 weeks. Coloured cells indicate active weeks for each task. M1 = Member 1 (green); M2 = Member 2 (blue). Section header rows are shown in dark blue.

Task	Owner	W1	W2	W3	W4	W5	W6	W7	W8
PHASE 1: Planning & Documentation									
Codebase review	M1+M2								
SRS document	M1								
Dev environment setup	M2								
Project plan & Gantt	M1+M2								
PHASE 2: Database & Core Architecture									
Finalize SQLAlchemy models	M1								
Alembic migration + room seeding	M2								
Auth blueprint	M1								
PHASE 3: Core Module Development									
Patient blueprint	M1								
Doctor blueprint	M1								
Nurse blueprint	M2								
Receptionist blueprint (full)	M2								
Store blueprint	M1								
PHASE 4: Frontend Templates									

Task	Owner	W1	W2	W3	W4	W5	W6	W7	W8
base.html + auth templates	M2								
Doctor & patient templates	M1								
Nurse templates	M2								
Receptionist & billing templates	M2								
Store templates	M1								
PHASE 5: Integration Testing									
RBAC and auth testing	M1								
Admission-to-discharge testing	M2								
Prescription & billing testing	M1+ M2								
Bug fixing & regression	M1+ M2								
PHASE 6: Documentation									
Design document	M2								
User manual	M2								
README + inline docs	M1								
PHASE 7: Deployment & Submission									
Production config (Gunicorn)	M1								
Final review & demo prep	M1+ M2								
Documentation packaging	M2								
Project submission	M1+ M2								

4. Resource Allocation

4.1 Effort Allocation by Phase

Phase	M1 Effort (hrs)	M2 Effort (hrs)	Total (hrs)	% of Project
Phase 1: Planning & Documentation	17	10	27	11%
Phase 2: Database & Core Architecture	19	9	28	11%
Phase 3: Core Module Development	54	54	108	43%
Phase 4: Frontend Templates	22	28	50	20%
Phase 5: Integration & Testing	20	20	40	16%
Phase 6: Additional Documentation	6	22	28	11%
Phase 7: Deployment & Submission	12	10	22	9%
TOTAL	150	153	303	100%

4.2 Weekly Load Summary

Week	Primary Focus	M1 Tasks	M2 Tasks	Est. Hours (M1)	Est. Hours (M2)
Week 1	Planning, Setup, SRS	SRS, models, project plan	Env setup, factory, migration	22	14
Week 2	Core Architecture	Auth, patient blueprint	Nurse blueprint, models	20	20
Week 3	Module Dev	Doctor blueprint, patient templates	Receptionist P1, base templates	20	20

Week	Primary Focus	M1 Tasks	M2 Tasks	Est. Hours (M1)	Est. Hours (M2)
Week 4	Module Dev + UI	Store blueprint, doctor UI	Receptionist P2, nurse UI	20	22
Week 5	UI + Testing	Store UI, RBAC testing	Receptionist UI, admission testing	18	20
Week 6	Testing & Bug Fixing	Prescription testing, bug fixes	Billing testing, user manual start	18	18
Week 7	Docs + Deployment	README, prod config, review	User manual, doc packaging	18	20
Week 8	Final Submission	Final review, submission	Final review, submission	14	19

5. Project Milestones

Milestone	Target Date	Description	Owner	Status
M1: Project Kick-off	Week 1, Day 1	Development environment configured; Git repository initialised; project plan agreed	M1 + M2	Planned
M2: SRS Submitted	Week 1, Day 7	Complete Software Requirements Specification document v1.0 submitted for review	M1	Planned
M3: Database Schema Stable	Week 2, Day 3	All SQLAlchemy models finalised; initial Alembic migration applied; rooms seeded	M1 + M2	Planned
M4: Authentication Live	Week 2, Day 7	Login, registration, RBAC routing functional; all role blueprints registered	M1	Planned
M5: Core Modules Complete	Week 4, Day 5	All 7 blueprints implemented and locally tested; prescription and billing logic working	M1 + M2	Planned
M6: Design Document Submitted	Week 3, Day 7	System Architecture and Design Document v1.0 submitted	M2	Planned
M7: All Templates Complete	Week 5, Day 2	All Jinja2 templates rendered correctly for all roles	M1 + M2	Planned
M8: Testing Complete	Week 6, Day 5	All integration tests passed; bug fix log finalised; regression suite clean	M1 + M2	Planned
M9: User Manual Delivered	Week 7, Day 3	Role-specific user manual with screenshots completed	M2	Planned
M10: Final Submission	Week 8, Day 5	All code, documents, and deployment config submitted in final package	M1 + M2	Planned

6. Risk Management

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R-01	Scope creep: additional features requested beyond SRS scope	Medium	High	Strictly follow SRS scope; defer out-of-scope features to a future version with change request documentation
R-02	Integration issues between blueprints (e.g., billing + admission data)	Medium	High	Conduct integration testing after each phase; use shared db.session to ensure transactional consistency
R-03	SQLite concurrency limitation in demo environment	Low	Medium	For demo, ensure single-user testing; document MySQL migration path for production scaling
R-04	Team member unavailability (illness, personal commitments)	Low	High	Cross-train both members on all modules; maintain shared Git repository; document all decisions in this plan
R-05	Database migration conflicts during parallel development	Medium	Medium	Coordinate model changes via Git; use feature branches; merge to main only after migration testing
R-06	Incorrect RBAC implementation leaving unauthorised access	Low	High	Code review all route files for RBAC guard at sprint end; add dedicated security test cases in Phase 5
R-07	Template rendering errors (Jinja2 variable mismatches)	Medium	Low	Test each template in isolation; use Flask debug mode to identify template errors quickly
R-08	Time overrun in Phase 3 (largest phase)	Medium	High	Built-in buffer of 2 days per week; if behind, deprioritise non-core features (e.g., advanced search) to protect core delivery

7. Communication and Collaboration Plan

7.1 Team Communication

Activity	Frequency	Participants	Medium	Purpose
Daily Stand-up	Daily (Mon-Fri)	M1 + M2	WhatsApp / in-person	Share progress, blockers, and next steps
Weekly Planning	Every Monday	M1 + M2	In-person / video call	Review previous week, plan current week's tasks
Code Review	Per feature completion	M1 + M2	GitHub Pull Request	Ensure code quality and catch integration issues
Supervisor Meeting	Weekly (as scheduled)	M1 + M2 + Supervisor	In-person	Progress review, guidance, and milestone sign-off

7.2 Version Control and Collaboration

- Git repository with a main branch for stable code and feature branches per task
- Commit messages follow the convention: [MODULE] Brief description (e.g., [AUTH] Add bcrypt password hashing)
- Pull requests required before merging to main; the other team member must review and approve
- Database migration files committed to the repository; never manually alter the schema
- All documents stored in a shared /Documentations/ folder in the repository and backed up to cloud storage

8. Quality Assurance Plan

8.1 Testing Strategy

Test Type	Scope	Responsible	Tools	Timing
Unit Testing	Individual route handlers and model methods (dispense, calculate_total, set_occupied)	M1 + M2	pytest + Flask test client	After each module in Phase 5
Integration Testing	Full workflow end-to-end: registration -> appointment -> admission -> discharge -> billing	M1 + M2	Flask test client + manual	Phase 5, Weeks 5-6
RBAC Testing	Verify each role can access only its permitted routes; test cross-role URL manipulation	M1	Manual + pytest	Week 5
Regression Testing	Full re-run of all test cases after bug fixes	M1 + M2	pytest test suite	Week 6 Day 3-5
User Acceptance Testing	Demonstrate all workflows to supervisor; validate against SRS requirements	M1 + M2	Live demo	Week 7

8.2 Code Quality Standards

- All route handlers include RBAC guard at the top
- All database writes use db.session.commit() within try-except blocks for error handling
- All forms use Flask-WTF with explicit validators; no raw request.form access without validation
- No plain-text passwords stored or logged at any point
- Python PEP 8 style guidelines followed throughout the codebase
- All significant functions documented with inline comments explaining business logic